



Chi-Squared Tests · Mark Scheme

Further Statistics 1 Chapter 6: Chi-Squared Tests. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Goodness-of-fit tests

1 (a) $6 - 1 = 5$ B1; (b) $5 - 1 - 1 = 3$ B1.

2 Each $E = 20$ B1; $X^2 = \frac{5^2 + 2^2 + 2^2 + 5^2}{20}$ M1; $= 2.9$ A1.

3 H_0 : the die is fair, H_1 : not fair; each $E = 20$ B1; $X^2 = \frac{25 + 4 + 4 + 36 + 36 + 25}{20} = 6.5$ M1; $6.5 < 11.070$: do not reject H_0 , no evidence the die is not fair A1.

4 Expected 90, 30, 30, 10 B1; $X^2 = \frac{64}{90} + \frac{25}{30} + \frac{9}{30} + 0$ M1; $= 1.844$ A1; $1.844 < 7.815$: do not reject H_0 ; the data are consistent with the 9 : 3 : 3 : 1 ratio A1.

5(a) H_0 : accidents are equally likely on each day (uniform), H_1 : not equally likely B1; each $E = \frac{120}{5} = 24$ B1.

5(b) $X^2 = \frac{36 + 36 + 16 + 49 + 121}{24}$ M1; $= 10.75$ A1; $\nu = 4$, and $10.75 > 9.488$ M1; reject H_0 : there is evidence that accidents are not equally likely on each day A1.

5(c) $10.75 < 13.277$, so at 1% H_0 would not be rejected: a different conclusion B1; the largest contributions are Friday (5.04), with more accidents than expected, and Thursday (2.04), with fewer B1.

Section B – Binomial and Poisson models

6 (a) $6 - 1 = 5$ B1; (b) $6 - 2 = 4$ B1.

7 Mean $= \frac{17 + 40 + 24}{50} = 1.62$, so $\hat{p} = \frac{1.62}{3} = 0.54$ B1; $50 \times 3 \times 0.54^2 \times 0.46$ M1; $= 20.12$ A1.

8 The expected frequencies are forced to agree with the data in one more way (e.g. the same mean), so one fewer frequency is free to vary B1 B1; small expected frequencies make $\frac{(O-E)^2}{E}$ unreliably large, so X^2 would not be well approximated by χ^2 B1.

9 H_0 : B(4, 0.7) is a suitable model; pool 0 and 1: $O = 18$, $E = 12.56$ B1; $X^2 = 2.3615 + 0.1344 + 0.7358 + 0.0286$ M1; $= 3.26$ A1; $\nu = 4 - 1 = 3$, $3.26 < 7.815$: do not reject H_0 , the claim is supported A1.

10(a) $\frac{18 + 40 + 48 + 24 + 30}{80} = \frac{160}{80} = 2$ B1.

10(b) $a = 80 \times \frac{e^{-2} 2^3}{3!} = 14.44$ B1; $b = 80 - (\text{sum of the others}) = 4.21$ B1.

10(c) H_0 : a Poisson distribution is a suitable model; $b < 5$, so pool $x = 4$ and $x \geq 5$: $O = 12$, $E = 11.43$ M1; $X^2 = 0.9300 + 0.6165 + 0.1263 + 0.1695 + 0.0284$ M1; $= 1.87$ A1; $\nu = 5 - 1 - 1 = 3$ B1; $1.87 < 7.815$: do not reject H_0 , the Poisson model is suitable A1.

Section C – Tests for association

11 $\nu = (3 - 1)(4 - 1) = 6$ B1; $E = \frac{60 \times 45}{200} = 13.5$ B1.

12 $E = \frac{\text{row total} \times \text{column total}}{100}$ M1; Year 12: 24, 24, 12 A1; Year 13: 16, 16, 8 A1.

13 $X^2 = \frac{36}{18} + \frac{0}{18} + \frac{36}{24} + \frac{36}{12} + \frac{0}{12} + \frac{36}{16}$ M1; $= 8.75$ A1; $8.75 > 5.991$: reject H_0 , there is evidence of an association between school and grade A1.

14 (a) The "Over 60" expected frequencies are all less than 5 B1; after combining the table is 2×3 , so $\nu = (2 - 1)(3 - 1) = 2$ B1. (b) $10.17 > 5.991$ M1; reject H_0 : there is evidence of an association between age group and method of travel A1.



15(a) H_0 : age group and preferred type of film are independent; H_1 : they are not independent *B1*.

15(b) $\frac{70 \times 80}{200} = 28$ *B1*; $\frac{65 \times 60}{200} = 19.5$ *B1*.

15(c) $p = \frac{10.5^2}{19.5} = 5.6538$ *B1*; $q = \frac{9.5^2}{19.5} = 4.6282$ *B1*.

15(d) $X^2 = 20.88$ *M1 A1*; $\nu = (3 - 1)(3 - 1) = 4$, $20.88 > 13.277$ *M1*; reject H_0 : there is evidence of an association between age group and preferred type of film *A1*.

15(e) The largest contributions show that 16–25s prefer action more than expected and drama less, while over 40s prefer drama more and action less *B1*.

Roar-level challenge

16 (a) $E = \frac{(a+b)(a+c)}{n}$, so $O - E = \frac{a(a+b+c+d) - (a+b)(a+c)}{n}$ *M1*; $= \frac{ad - bc}{n}$ *A1*; each row and column total of $O - E$ is zero, so the four values are $\pm \frac{ad - bc}{n}$ *B1*. (b) $|O - E| = \frac{1200 - 200}{100} = 10$, with $E = 20, 30, 20, 30$ *M1*; $X^2 = 100 \left(\frac{1}{20} + \frac{1}{30} + \frac{1}{20} + \frac{1}{30} \right) = \frac{50}{3} = 16.67$ *A1*.